EB-JEPA: Energy-Based Joint Embedding Predictive Architecture for Self-Supervised EEG Representation Learning

A Technical Report on the TUAB Abnormality Benchmark

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Abstract

We study self-supervised representation learning for clinical EEG through an Energy-Based Joint Embedding Predictive Architecture (EB-JEPA). Two independently corrupted views of the same 10-second EEG window are aligned in latent space by an energy function combining invariance, variance, and covariance regularisation (VICReg [2]) or a stronger Gaussianisation surrogate (SIGReg / Batched Characteristic Slicing [4]). Four encoder families are evaluated on the TUH Abnormal EEG Corpus (TUAB v3.0.1 [11], 2717 training and 276 evaluation recordings, binary normal/abnormal classification): a temporal Conv1D network (0.4M parameters), a patch Transformer in the style of LaBraM [5] (1.2M), an FFT-tokenised Transformer in the style of BIOT [18] (1.2M), and a hierarchical channel-pool Transformer in the style of EEGPT [16] (0.8M).

Our best frozen encoder (Conv1D + corruption augmentation + SIGReg) reaches **0.775 balanced accuracy (BACC) per-window** and **0.825 BACC per-recording** (mean-pooling 16 window embeddings per patient), matching ContraWR and approaching LaBraM-Base (0.814 per-window). Fine-tuning adds 1.2 pp to reach **0.837 BACC per-recording** and AUROC **0.919**.

The central contribution of this report is *diagnostic*: we identify and document three systematic failure modes — a VICReg coefficient bug (λ=1 vs. the paper-correct λ=25), an architectural conflict between EEGPT's channel-pooling and channel-drop augmentation, and a per-window vs. per-recording evaluation gap ($\approx$5 pp for the same encoder) that initially caused an inflated performance claim. We describe how each failure was diagnosed, quantified, and corrected. An auxiliary spectral regularisation loss further yields the strongest per-recording frozen result (0.836 BACC), leaving open whether it provides independent signal or compensates for the underweighted invariance term.

Contents

Introduction

This report documents the application of Energy-Based Joint Embedding Predictive Architectures (EB-JEPA, [14]) to clinical EEG abnormality detection on the TUH Abnormal EEG Corpus (TUAB). The work was conducted during a 48-hour hackathon at the Dalia cluster, under the broader EB-JEPA research initiative.

Structure of the report.

Section 1: State of the Art. We first clarify the three distinct *scoring styles* used across the EEG literature (per-window, per-recording via window aggregation, per-recording from full recording), as confusing them produces spurious comparison advantages. We then provide the complete *provenance table* of published results and our own ground-truth numbers from the repository, covering supervised ConvNets, masked SSL (BIOT, LaBraM), contrastive SSL (BENDR, ContraWR), and recent foundation models (FEMBA [13], EEG-Bench [6]).

Section 2: EB-JEPA Results. We present the two-view energy concept underlying EB-JEPA, describe the TUAB dataset and the critical distinction between SSL training and probe evaluation splits, and then report all experimental results from the hackathon in a single comprehensive table.

Section 3: Per-Model Mathematical Descriptions. For every encoder family tested, we provide the complete energy function with explicit loss weights, encoder architecture details, and the MLP probe specification. We then describe the training and evaluation protocol exactly as implemented.

Section 4: Combined Best Results. We consolidate the best numbers from the literature and from this work, filtered to per-recording metrics only (the clinically meaningful unit of analysis for TUAB).

Section 5: Conclusion and Future Work. We conclude with open scientific questions, methodological lessons, and a prioritised list of future experiments.

Research question. Can a small (0.4M parameter) encoder pre-trained *without labels* on 2717 unlabelled EEG recordings learn representations that generalise to held-out patients and match or exceed supervised models trained with labels?

1 State of the Art on TUAB EEG Abnormality Detection

1.1 Scoring Styles

Published results on TUAB use heterogeneous evaluation conventions. We precisely define four *scoring styles* used in the literature:

(S1) Per-window (WIN). The evaluation set is segmented into all non-overlapping 10-second windows. Each window is independently classified; it inherits the label of its parent recording. BACC is computed over all individual window predictions. For TUAB eval (276 recordings, 30–120 min each), this yields ≈400k test samples. This is the convention used by BIOT [18], LaBraM [5], ContraWR [17], and most published SSL models. The test samples are *not independent*: windows from the same recording share the same label and much of the same signal, inflating effective sample size.

(S2) Per-recording via window embedding aggregation (REC-WIN). A fixed number N of evenly-spaced windows are extracted per recording. Each window is encoded separately; the N embeddings are *mean-pooled* into one vector. The probe classifies this pooled vector; one prediction is made per recording. This is the default mode of our framework with $N=16$. The noise-averaging effect of pooling systematically raises BACC by ≈5 pp over WIN for the same encoder and probe.

(S3) Per-recording via majority vote (REC-VOTE). Similar to REC-WIN but aggregates *decisions* rather than embeddings: each window is independently classified, and the recording label is the majority class. This style is used in some clinical pipelines but not in the SSL literature surveyed here.

(S4) Per-recording from the full recording (REC-FULL). The entire recording (variable length, up to 120 min) is fed as a single input to a model capable of handling variable-length sequences (e.g., an SSM or a hierarchical model). No windowing or aggregation is required. None of our models use this mode; it is listed for completeness.

Critical warning. Comparing a REC-WIN or REC-FULL number directly to a WIN number as if they were the same quantity produces a spurious ≈5 pp advantage to the per-recording model. Section 5 describes how this gap was inadvertently exploited and then corrected in this project.

1.2 State-of-the-Art Provenance Table

Table 1 compiles every TUAB result we could verify against a primary source. We call this the *provenance table*: every entry is annotated with its source paper, so the reader can verify the number directly. Entries in yellow denote results from the EEG-Bench benchmark [6] or FEMBA [13] added in this update.

Key observations from Table 1.

1. The best published TUAB BACC is held by FEMBA-Huge at 81.82% (fine-tuned, trained on 21,000h of unlabelled EEG from TUEG), followed by LaBraM-Large/Huge (82.26–82.58% in FEMBA's evaluation; 83.8% fine-tuned in EEG-Bench).
2. Among methods using the same TUAB training set without external pretraining, LaBraM-Base (0.814 frozen, WIN) is the primary target for our approach.
3. Our best frozen WIN BACC (0.775, SIGReg+corrupt) matches ContraWR and falls 3.9 pp below LaBraM-Base, with only 0.4M parameters and 20 training epochs.
4. The spectral audit of Bindra & Panwar [3] reveals that supervised and SSL models on TUAB rely substantially on the aperiodic (1/f) broadband spectral slope (ΔBACC 0.07–0.13 after flattening), which may partially confound abnormality detection with age-related spectral changes. EB-JEPA's SIGReg Gaussianisation surrogate is architecturally designed to prevent representational collapse, but whether it avoids aperiodic shortcuts remains an open question.

2 EB-JEPA Results

2.1 EB-JEPA Two-View Energy Concept

EB-JEPA is our EEG adaptation of the Joint Embedding Predictive Architecture paradigm [1, 10]. We adopt a *two-view energy-based* formulation adapted to time-series signals, which differs from the standard image JEPA in that both views are corrupted versions of the *same* window (rather than masked spatial patches), and the encoder is trained to produce the same latent representation for both views.

Table 1: Provenance table: state-of-the-art results on TUAB EEG abnormality detection. WIN = per-window; REC-WIN = per-recording via window embedding pooling; REC-FT = per-recording, fine-tuned end-to-end. "Frozen" = frozen encoder + linear/MLP probe. "FT" = fine-tuned encoder. [a]Accuracy (not BACC), early eval script. [b]Reported as accuracy; TUAB eval split is near-balanced (54% normal, 46% abnormal), so acc ≈ BACC within 2 pp. [c]AUROC estimated from ROC curve in the paper. [d]Protocol unclear; likely REC-WIN with averaging over non-overlapping chunks before classification. ⋆AUROC measured under REC-WIN (N=16); WIN AUROC not recorded. VICReg* = buggy `inv_coeff=1`; VICReg = corrected `inv_coeff=25`.

Scoring	Model	BACC	AUROC	Params	Source	Mode
Floor						
WIN	Random encoder	0.52	–	–	This work	Frozen
Supervised baselines						
WIN	ShallowConvNet [12]	0.777	0.857	44k	Schirrmeister 2017	FT
WIN	EEGNet [9]	0.796	0.882	9k	Kiessner 2023 [7]	FT
WIN	Deep4Net [12]	0.816[b]	–	–	Schirrmeister 2018	FT
Self-supervised foundation models (published)						
WIN	ContraWR [17]	0.775	–	–	Yang 2021	Frozen
WIN	BENDR [8]	0.717	–	4M	Kastrati 2025 [6]	FT[d]
WIN	BIOT [18]	0.802	–	3.2M	Yang 2023	Frozen
WIN	BIOT [18]	0.802[c]	–	3.2M	Bindra 2026 [3]	FT[d]
WIN	LaBraM-Base [5]	0.814	0.890[c]	5.8M	Jiang 2024	Frozen
WIN	LaBraM-Base [5]	0.786	–	5.8M	Bindra 2026 [3]	FT[d]
REC-FT	LaBraM-Base [5]	0.838	–	5.8M	Kastrati 2025 [6]	FT
REC-FT	Neuro-GPT [6]	0.696	–	–	Kastrati 2025 [6]	FT
WIN	EEGPT [16]	0.801	–	10M	Bindra 2026 [3]	FT[d]
REC-FT	FEMBA-Base [13]	0.8105	0.8894	47.7M	Tegon 2025	FT
REC-FT	FEMBA-Large [13]	0.8147	0.8892	77.8M	Tegon 2025	FT
REC-FT	FEMBA-Huge [13]	**0.8182**	**0.9005**	386M	Tegon 2025	FT
EB-JEPA (this work) – *per-window (WIN), frozen encoder*						
WIN	Conv1D + VICReg*	0.756	0.888⋆	0.4M	This work	Frozen
WIN	Conv1D + VICReg* + corrupt	0.770	0.904⋆	0.4M	This work	Frozen
WIN	Conv1D + SIGReg + corrupt	0.775	0.904⋆	0.4M	This work	Frozen
WIN	Conv1D + VICReg* + spectral 0.1	0.765	0.887⋆	0.4M	This work	Frozen
WIN	LaBraM-style + VICReg*	0.775[a]	–	1.2M	This work	Frozen
WIN	BIOT-style + VICReg*	0.775[a]	–	1.2M	This work	Frozen
WIN	BIOT-style + SIGReg	0.783[a]	–	1.2M	This work	Frozen
WIN	EEGPT-style + VICReg*	collapsed	–	0.8M	This work	Frozen
EB-JEPA (this work) – *per-recording via window pooling (REC-WIN, N=16)*						
REC-WIN	Conv1D + VICReg*	0.796	0.888	0.4M	This work	Frozen
REC-WIN	Conv1D + VICReg* + corrupt	0.825	0.904	0.4M	This work	Frozen
REC-WIN	Conv1D + SIGReg + corrupt	0.825	0.904	0.4M	This work	Frozen
REC-WIN	Conv1D + VICReg* + spectral 0.1	**0.836**	0.887	0.4M	This work	Frozen
REC-WIN	Conv1D + VICReg* + corrupt (FT)	0.812	**0.908**	0.4M	This work	FT

Two-view concept with two corruption pathways. Given a 10-second EEG window $\mathbf{x} \in \mathbb{R}^{C \times T}$ (C=19 channels, T=2000 samples at 200 Hz), two independently corrupted views are produced as follows:

- **Corruption pathway (a):** Apply augmentation stack $\mathcal{A}_a$ to $\mathbf{x}$ to obtain $\tilde{\mathbf{x}}_a = \mathcal{A}_a(\mathbf{x})$.
- **Corruption pathway (b):** Apply independently sampled augmentation stack $\mathcal{A}_b$ to obtain $\tilde{\mathbf{x}}_b = \mathcal{A}_b(\mathbf{x})$.

Both $\mathcal{A}_a$ and $\mathcal{A}_b$ are drawn from the same family but with independently sampled random parameters (noise seed, masked channels, etc.). A shared encoder f_θ maps each view to a latent vector:

$$\mathbf{z}_a = g_\phi(f_\theta(\tilde{\mathbf{x}}_a)), \quad \mathbf{z}_b = g_\phi(f_\theta(\tilde{\mathbf{x}}_b)), \quad \mathbf{z}_a, \mathbf{z}_b \in \mathbb{R}^{d_p},\ d_p{=}1024, \tag{1}$$

where g_ϕ is a projector MLP ($d \to 1024 \to 1024$) that maps from the encoder output dimension d=256 to the projector space. The energy function $E(\mathbf{z}_a, \mathbf{z}_b) = \|\mathbf{z}_a - \mathbf{z}_b\|^2$ penalises disagreement between the two views. Without an anti-collapse mechanism, the trivial global minimum $\mathbf{z}_a = \mathbf{z}_b = \mathbf{0}$ satisfies E=0 with zero information. VICReg [2] and SIGReg prevent this collapse by adding variance and covariance regularisation terms (described formally in Section 3.5).

The two-view concept is illustrated in Figure 1: the world model (encoder) is trained to map two different corruptions of the same EEG brain state to the same point in latent space.

$$\mathbf{x} \in \mathbb{R}^{19 \times 2000} \text{ (EEG window, 10s)}$$

$$\downarrow \quad \mathcal{A}_a(\cdot) \qquad\qquad \mathcal{A}_b(\cdot) \quad \downarrow$$

$$\tilde{\mathbf{x}}_a \xrightarrow{f_\theta \circ g_\phi} \mathbf{z}_a \in \mathbb{R}^{1024} \qquad \mathbf{z}_b \in \mathbb{R}^{1024} \xleftarrow{f_\theta \circ g_\phi} \tilde{\mathbf{x}}_b$$

$$E = \underbrace{\|\mathbf{z}_a - \mathbf{z}_b\|^2}_{\text{invariance (energy)}} + w_V \underbrace{L_V(\mathbf{z}_a, \mathbf{z}_b)}_{\text{variance (anti-collapse)}} + w_C \underbrace{L_C(\mathbf{z}_a, \mathbf{z}_b)}_{\text{covariance (decorrelation)}}$$

Figure 1: EB-JEPA two-view training scheme. Two independently corrupted views of the same EEG window (pathways a and b) are mapped to the same latent space through a shared encoder f_θ and projector g_ϕ. The energy function combines invariance with variance and covariance regularisation to prevent representational collapse. The encoder acts as the *world model*: it learns to abstract away the specific corruption and retain the clinically relevant brain state.

2.2 EEG Dataset: SSL Training vs. Probe Evaluation Splits

All experiments use the **TUH Abnormal EEG Corpus (TUAB) v3.0.1** [11]. A critical distinction exists between the *SSL split* (used during pre-training) and the *probe split* (used to train and evaluate the downstream classifier).

Recording configuration. 19-channel scalp EEG in the international 10-20 system, sampled at 256 Hz and resampled to 200 Hz. Recordings span 10–120 minutes (clinical monitoring sessions). The binary label (*normal* vs. *abnormal*) is assigned by a board-certified neurologist to the entire recording.

Patient-disjoint splits. TUAB provides an official patient-disjoint train/eval split:

- **SSL training split** (2 717 recordings: 1 385 normal, 1 332 abnormal): used for SSL pre-training *without labels*. The encoder f_θ sees only the raw EEG signal; the normal/abnormal labels are discarded.
- **Evaluation split** (276 recordings: 150 normal, 126 abnormal): used exclusively for final performance estimation. No hyperparameter selection is performed on this split.

No recording from a given patient appears in both splits, preventing patient-level data leakage.

SSL pre-training (ssl split). The encoder is pre-trained using the SSL training split. Labels are discarded. The two-view SSL objective (VICReg or SIGReg) is optimised over 20 epochs on non-overlapping 10-second windows extracted from all 2 717 training recordings.

Probe training (probe split = ssl training split with labels). After SSL pre-training, the encoder is frozen. An MLP probe is trained on the *same* 2 717 training recordings, now using the normal/abnormal labels. For each recording, N=16 evenly-spaced 10-second windows are extracted and encoded; their embeddings are mean-pooled into one 256-dim vector (REC-WIN protocol). The probe is trained on these pooled vectors.

Comparison with the published SSL protocol. The published LaBraM, BIOT, and ContraWR evaluations use the per-window (WIN) protocol on a larger eval split (≈2339 recordings, ≈409k test windows). Our eval split (276 recordings) is the official TUAB v3.0.1 split. The different eval set sizes contribute to difficulty in direct comparison, beyond the WIN vs. REC-WIN protocol gap.

2.3 Full Results Table for JEPA Models Tested

Table 2 reports all encoder × loss × augmentation combinations evaluated during the hackathon.

Key observations.

1. **Conv1D dominates** all encoder families despite fewest parameters (0.4M).
2. **Corruption augmentation** (time masking + spike injection) is the single most impactful addition: +0.014 WIN BACC, +0.029 REC BACC.
3. **Spectral regularisation at coefficient 0.1** gives the best per-recording frozen result (0.836) but was trained with the VICReg bug; result must be interpreted with caution.
4. **LaBraM-style and BIOT-style** reach the same accuracy (0.775) despite fundamentally different tokenisation (raw patches vs. FFT magnitude), suggesting the bottleneck is encoder capacity at 1.2M params, not tokenisation.
5. **EEGPT-style collapsed** under both loss functions with the buggy coefficients; rehabilitation with fixed VICReg was pending at write-up time.
6. **SIGReg ≥ VICReg** for Conv (+0.005 WIN), but **SIGReg < VICReg** for LaBraM-style (−0.050 accuracy) — the loss–architecture interaction is non-trivial.
7. **Predictive EMA-JEPA** (GRU predictor, frame-prediction objective) achieved only 0.764 REC BACC — the *worst* result of all variants. The frame-prediction objective encourages temporal dynamics representations rather than class-discriminative statistics.

3 Per-Model Mathematical Descriptions

For each encoder, we provide: (i) the complete energy function E with explicit loss component definitions and weight choices; and (ii) the encoder architecture with its primary design choices.

Table 2: Complete EB-JEPA results on TUAB v3.0.1 (276-recording eval split). “Frozen” = frozen encoder, MLP probe. “FT” = entire network fine-tuned during probe training. WIN = per-window BACC; REC = per-recording BACC (N=16 windows, mean-pooled). “collapsed” = BACC $\leq$ random init floor. [a]Accuracy (not BACC), early eval script. ⋆AUROC under REC-WIN; WIN AUROC not recorded. VICReg* = buggy `inv_coeff=1`; VICReg = corrected `inv_coeff=25`. FT rows are 3-seed means at the *final* epoch (no best-epoch selection).

Model / Config	**BACC$_{\text{WIN}}$**	**BACC$_{\text{REC}}$**	**AUROC$_{\text{REC}}$**	**Score style**	**Probe**	**SSL loss**	**Data**
Conv1D encoder (0.4M parameters) – spectral regularisation ablation							
Conv + VICReg* (base)	0.756	0.796	0.888	WIN / REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + spec 0.05	–	0.821	0.884	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + spec 0.1	0.765	0.836	0.887	WIN / REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + spec 0.3	–	0.789	0.877	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + spec 1.0	–	0.785	0.874	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + fftc 0.05	–	0.807	0.887	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + fftc 0.1	–	0.798	0.881	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + fftc 0.3	–	0.802	0.889	REC-WIN	Frozen	VICReg*	SSL
Conv1D encoder – corruption augmentation ablation (3 seeds)							
Conv + VICReg* + corrupt (s0)	0.770	0.825	0.904	WIN / REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + corrupt (s1)	–	0.816	0.891	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + corrupt (s2)	–	0.818	0.906	REC-WIN	Frozen	VICReg*	SSL
Conv + VICReg* + corrupt (big)	–	0.805	0.892	REC-WIN	Frozen	VICReg*	SSL
Conv + corrupt + spec 0.1	–	0.802	0.899	REC-WIN	Frozen	VICReg*	SSL
Conv + SIGReg + corrupt	0.775	0.825	0.904	WIN / REC-WIN	Frozen	SIGReg	SSL
Conv1D encoder – fine-tuning and multi-corpus							
Conv + VICReg* + corrupt (FT)	–	0.812	0.908	REC-WIN	Fine-tuned	VICReg*	SSL
Conv + VICReg* + multicorpus	–	0.812	–	REC-WIN	Frozen	VICReg*	SSL
Conv + multicorpus (FT)	–	0.812	–	REC-WIN	Fine-tuned	VICReg*	SSL
EMA-JEPA on Conv (150ep)	–	0.764	–	REC-WIN	Frozen	Pred-JEPA	SSL
LaBraM-style patch Transformer (1.2M parameters)							
LaBraM-style + VICReg*	0.775[a]	–	–	WIN	Frozen	VICReg*	SSL
LaBraM-style + SIGReg	0.725[a]	–	–	WIN	Frozen	SIGReg	SSL
BIOT-style FFT Transformer (1.2M parameters)							
BIOT-style + VICReg*	0.775[a]	–	–	WIN	Frozen	VICReg*	SSL
BIOT-style + SIGReg	0.783[a]	–	–	WIN	Frozen	SIGReg	SSL
EEGPT-style channel-pool Transformer (0.8M parameters)							
EEGPT-style + VICReg*	collapsed	–	–	WIN	Frozen	VICReg*	SSL
EEGPT-style + SIGReg	collapsed	–	–	WIN	Frozen	SIGReg	SSL
EEGPT-style + VICReg (fixed, ep.11)	pending	–	–	–	–	VICReg	SSL

“Data” column: SSL = TUAB SSL training split (2717 recordings, no labels). Multicorpus = TUAB+TUEV+TUSZ+TUEP (4× data).

3.1 Conv1D Temporal Encoder

Energy function.

$$\boxed{E_{\text{Conv}} = w_2 \cdot L_2 + w_V \cdot L_V + w_C \cdot L_C + w_{\text{spec}} \cdot L_{\text{spec}}} \tag{2}$$

with weights $(w_2, w_V, w_C) = (25, 25, 1)$ for VICReg and $w_{\text{spec}} \in \{0, 0.05, 0.1, 0.3, 1.0\}$ for the spectral ablation ($0 =$ no spectral loss).

Component L_2 (invariance / reconstruction energy):

$$L_2 = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{z}_{a,i} - \mathbf{z}_{b,i}\|^2, \tag{3}$$

where $\mathbf{z}_{a,i}, \mathbf{z}_{b,i} \in \mathbb{R}^{d_p}$ are the projected embeddings of the two views for sample i in the batch of size N. This is the core energy: minimising L_2 pushes both views to the same point.

Component L_V (variance / anti-collapse hinge):

$$L_V = \frac{1}{2d_p} \sum_{j=1}^{d_p} \Big[\max\big(0,\ \gamma - \text{Std}(\mathbf{Z}_{a,:,j})\big) + \max\big(0,\ \gamma - \text{Std}(\mathbf{Z}_{b,:,j})\big)\Big], \quad \gamma = 1. \tag{4}$$

$\mathbf{Z}_a \in \mathbb{R}^{N \times d_p}$ is the matrix of all projected embeddings for view a; $\mathbf{Z}_{a,:,j}$ is its j-th column. L_V penalises any dimension j whose standard deviation falls below 1, preventing collapse to a constant vector.

Component L_C (covariance / decorrelation):

$$L_C = \frac{1}{2d_p(d_p-1)} \Big[\sum_{i \neq j} \text{Cov}(\mathbf{Z}_a)_{ij}^2 + \sum_{i \neq j} \text{Cov}(\mathbf{Z}_b)_{ij}^2\Big], \tag{5}$$

where $\text{Cov}(\mathbf{Z}) = (\mathbf{Z} - \bar{\mathbf{Z}})^\top (\mathbf{Z} - \bar{\mathbf{Z}})/(N-1) \in \mathbb{R}^{d_p \times d_p}$. L_C penalises correlated dimensions, encouraging the encoder to spread information across independent dimensions.

Component L_{spec} (spectral auxiliary):

$$L_{\text{spec}} = \frac{1}{Nf} \sum_{i=1}^{N} \sum_{m=1}^{f} \Big(\log \hat{S}_{i,m} - \log S_{i,m}\Big)^2, \tag{6}$$

where $S_{i,m}$ is the target Welch power spectral density at frequency bin m for sample i, and $\hat{S}_{i,m}$ is predicted by an auxiliary linear head from the encoder output $\mathbf{z}_i \in \mathbb{R}^d$. This DDSP-inspired loss encourages the encoder to capture spectral structure.

Architecture. The Conv1D encoder (from scratch, this work) processes raw EEG $\mathbf{X} \in \mathbb{R}^{C \times T}$, C=19, T=2000.

$$\mathbf{H}^{(l)} = \text{GELU}\Big(\text{BN}\big(\text{Conv1d}_{F_l,\, k=7,\, s=2}(\mathbf{H}^{(l-1)})\big)\Big), \quad l = 1, \ldots, 4, \tag{7}$$

with $F_0 = 19$ (input channels), $F_1 = 64$, $F_2 = 128$, $F_3 = 256$, $F_4 = 256$. Each stride-2 convolution halves the time dimension: $T \to 1000 \to 500 \to 250 \to 125$. Global average pooling over the time dimension, followed by a linear projection $256 \to 256$:

$$\mathbf{z} = \mathbf{W} \left(\frac{1}{125} \sum_{t=1}^{125} \mathbf{H}^{(4)}_{:,t} \right) \in \mathbb{R}^{256}. \tag{8}$$

Total parameters: ≈ 0.4M. Projector MLP: $256 \to 1024 \to 1024$ (non-linear, batch-normed), output $\mathbf{z} \mapsto \hat{\mathbf{z}} \in \mathbb{R}^{1024}$.

3.2 LaBraM-Style Patch Transformer Encoder

Energy function. Same as Conv1D: $E_{\text{LaBraM}} = 25L_2 + 25L_V + L_C$ (or SIGReg: $E = L_2 + \lambda_{\text{SIG}} L_{\text{SIG}}$, see Section 3.5).

Architecture. Inspired by LaBraM [5] without the VQ-codebook neural tokenizer (from scratch in this work). Each EEG channel is split into P=10 non-overlapping patches of w=200 timepoints (1 second each).

$$\mathbf{p}_{c,k} = \mathbf{X}_{c,\,(k-1)w:kw} \in \mathbb{R}^w, \quad c \in [C],\ k \in [P]. \tag{9}$$

Linear patch embedding: $\mathbf{e}_{c,k} = \mathbf{W}_p \mathbf{p}_{c,k} + \mathbf{b}_p \in \mathbb{R}^{d_e}$, d_e=128. Learnable temporal embedding $\mathbf{te}_k \in \mathbb{R}^{d_e}$ and channel (spatial) embedding $\mathbf{se}_c \in \mathbb{R}^{d_e}$:

$$\hat{\mathbf{e}}_{c,k} = \mathbf{e}_{c,k} + \mathbf{te}_k + \mathbf{se}_c. \tag{10}$$

Sequence of $C \times P = 190$ tokens fed to 4 Transformer blocks [15] with pre-layer normalisation:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{\text{LN}(Q)\,\text{LN}(K)^\top}{\sqrt{d_h}}\right) V, \quad d_h = d_e/n_h = 32,\ n_h\text{=}4. \tag{11}$$

FFN hidden dimension: $4d_e = 512$. Mean-pool over all 190 tokens: $\bar{\mathbf{e}} = \frac{1}{190}\sum \hat{\mathbf{e}}$. Linear projection $d_e \to 256$: $\mathbf{z} = \mathbf{W}_{\text{out}}\bar{\mathbf{e}}$. Total parameters: ≈ 1.2M.

3.3 BIOT-Style FFT Transformer Encoder

Energy function. Same as Conv1D (VICReg or SIGReg).

Architecture. Inspired by BIOT [18] (from scratch in this work). The key difference from LaBraM-style is the tokenisation: instead of raw time samples, each patch $\mathbf{p}_{c,k}$ is embedded via its Fourier magnitude spectrum.

$$\mathbf{s}_{c,k}[m] = \left| \sum_{n=0}^{w-1} \mathbf{p}_{c,k}[n]\, e^{-j2\pi mn/w} \right|, \quad m = 0, \ldots, \tfrac{w}{2}, \quad w\text{=}200. \tag{12}$$

This yields a 101-dimensional frequency-domain feature vector per patch (the one-sided magnitude spectrum). Linear projection: $\mathbf{e}_{c,k} = \mathbf{W}_s \mathbf{s}_{c,k} \in \mathbb{R}^{d_e}$, d_e=128. The remaining architecture (channel embedding, position embedding, Transformer) is identical to LaBraM-style. The FFT discards phase information, which is why BIOT-style underperforms on tasks where temporal dynamics (e.g., spike-wave complexes, burst suppression) are diagnostically relevant. Total parameters: ≈ 1.2M.

3.4 EEGPT-Style Hierarchical Channel-Pool Transformer

Energy function. Same as Conv1D (VICReg or SIGReg).

Architecture. Inspired by EEGPT [16] (from scratch in this work). The key architectural feature is a *spatial channel-pooling step* before the temporal Transformer.

Each patch $\mathbf{p}_{c,k} \in \mathbb{R}^w$ (w=200) is linearly embedded with a learnable channel codebook $\mathbf{s}_c$:

$$\text{tok}_{c,k} = \mathbf{W}_p \mathbf{p}_{c,k} + \mathbf{s}_c \in \mathbb{R}^{d_e}, \quad d_e\text{=}128. \tag{13}$$

Spatial channel-pooling (cross-attention). A learnable query $\mathbf{q} \in \mathbb{R}^{1\times d_e}$ attends over all C=19 channel tokens at time patch k:

$$\mathbf{g}_k = \mathrm{Attn}(\mathbf{q};\ [\mathrm{tok}_{1,k}, \ldots, \mathrm{tok}_{C,k}]) \in \mathbb{R}^{d_e}. \tag{14}$$

This reduces the $C \times P = 190$ token sequence to P=10 summary tokens.

Temporal Transformer. 4 standard Transformer blocks with learned positional embeddings over the 10 summary tokens: $\mathbf{h} = \mathrm{TransformerEncoder}([\mathbf{g}_1, \ldots, \mathbf{g}_P]) \in \mathbb{R}^{P\times d_e}$. Mean-pool and linear projection $d_e \to 256$: $\mathbf{z} \in \mathbb{R}^{256}$. Total parameters: ≈ 0.8M.

Failure mode: channel-drop vs. channel-pool conflict. The channel-pool cross-attention maps 19 channel tokens to 1 summary token per time patch. When the *channel-drop augmentation* (p=0.2 per channel) independently zeros different channels in view a and view b, the resulting summary tokens $\mathbf{g}_k^a$ and $\mathbf{g}_k^b$ look structurally different even though they derive from the same underlying signal. The invariance loss $L_2 = \|\mathbf{z}_a - \mathbf{z}_b\|^2$ cannot be minimised without the encoder learning channel-invariant features, which defeats the purpose of the channel embedding. This created the EEGPT collapse observed in our experiments.

3.5 SSL Objectives: VICReg and SIGReg

VICReg [2]. Let $\mathbf{Z}, \mathbf{Z}' \in \mathbb{R}^{N\times d_p}$ be the batch of projected embeddings for views a and b.

$$\boxed{E_{\mathrm{VICReg}} = w_2 \cdot L_2 + w_V \cdot L_V + w_C \cdot L_C} \quad \text{with } (w_2, w_V, w_C) = (25, 25, 1). \tag{15}$$

L_2, L_V, L_C are defined in Section 3.1. The paper-correct coefficients $(25, 25, 1)$ from Bardes et al. [2] were used in all fixed runs; the buggy runs used $(1, 1, 1)$ (see Section 5).

SIGReg (Batched Characteristic Slicing / BCS). SIGReg replaces L_V and L_C with a Gaussianisation surrogate based on the Epps–Pulley test statistic [4]. For K random unit-sphere projection directions $\mathbf{v}_k \sim \mathcal{U}(\mathcal{S}^{d_p-1})$:

$$L_{\mathrm{SIG}} = \frac{1}{2K}\sum_{k=1}^{K}\Big[\mathrm{EP}_N\big(\{\mathbf{v}_k^\top \mathbf{z}_{a,i}\}_i\big) + \mathrm{EP}_N\big(\{\mathbf{v}_k^\top \mathbf{z}_{b,i}\}_i\big)\Big], \tag{16}$$

where $\mathrm{EP}_N(\{u_i\})$ is the Epps–Pulley test statistic measuring deviation of the empirical distribution from Gaussian:

$$\mathrm{EP}_N(\{u_i\}) = \int_{-3}^{3} e^{-t^2/2} \left|\hat{\varphi}_N(t) - e^{-t^2/2}\right|^2 dt, \tag{17}$$

where $\hat{\varphi}_N(t) = \frac{1}{N}\sum_{i=1}^{N} e^{jtu_i}$ is the empirical characteristic function. In practice, the integral is approximated by 10 equally-spaced points in $[-3, 3]$. The full energy is:

$$\boxed{E_{\mathrm{SIGReg}} = L_2 + \lambda \cdot L_{\mathrm{SIG}}} \quad \text{with } \lambda = 10,\ K\text{=}256. \tag{18}$$

SIGReg is theoretically stronger than VICReg's variance hinge: representations collapse iff their random projections become non-Gaussian, which is a more comprehensive collapse test. In practice, SIGReg improves Conv1D by 0.005 WIN BACC but *hurts* LaBraM-style (-0.050 accuracy), showing that the loss–architecture coupling is non-trivial.

3.6 MLP Classifier Architecture

The downstream classification probe is a 3-layer MLP:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1), \quad \mathbf{W}_1 \in \mathbb{R}^{128\times 256}, \tag{19}$$

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2), \quad \mathbf{W}_2 \in \mathbb{R}^{64\times 128}, \tag{20}$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3), \quad \mathbf{W}_3 \in \mathbb{R}^{2\times 64}. \tag{21}$$

Implemented via `sklearn.neural_network.MLPClassifier` with `hidden_layer_sizes=(128, 64)`, early stopping, and ℓ_2 regularisation.

In **frozen mode**, only $\{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^{3}$ are updated; f_θ parameters receive no gradient. In **fine-tuned mode**, all parameters (encoder + probe) are optimised jointly with a lower learning rate (10^{-5}) for the encoder.

Several probe architectures were considered but not implemented: logistic regression (sklearn LogReg), which is the standard SSL community probe and would allow direct comparison to SimCLR/MoCo literature, was never added. Our MLP is more generous than LogReg, which inflates absolute numbers relative to published comparisons.

3.7 Training and Testing Protocol

SSL pre-training (ssl split).

- Dataset: TUAB SSL training split (2 717 recordings, labels discarded).
- Window extraction: non-overlapping 10-second windows, 2 000 timepoints at 200 Hz; per-channel z-score normalisation.
- Optimiser: AdamW, β_1=0.9, β_2=0.999, weight decay 10^{-4}, learning rate 10^{-4} with cosine annealing.
- Batch size: 64 windows (128 views per step after two-view augmentation).
- Duration: 20 epochs over all training windows.
- Augmentation stack (applied independently to each view): Gaussian noise (σ=0.1), amplitude jitter (±20%), channel dropout (20% of channels zeroed independently per view), time masking (20% of timepoints zeroed, corruption variant), spike injection (±6σ, 20% of windows, corruption variant).

Probe training (ssl training split with labels).

- Dataset: 2 717 TUAB training recordings with normal/abnormal labels.
- Feature extraction: N=16 evenly-spaced 10-second windows per recording → frozen encoder → 16 embeddings of $\mathbb{R}^{256}$ → mean-pool → one 256-dim vector per recording.
- Probe: sklearn MLPClassifier, `hidden_layer_sizes=(128, 64)`, `max_iter=500`, `early_stopping=True`
- Standardisation: StandardScaler fitted on training embeddings only; same scaler applied to eval embeddings.

Evaluation (eval split).

- Dataset: 276 TUAB evaluation recordings (patient-disjoint from training).
- **Per-window (WIN):** extract all non-overlapping 10-second windows, classify each independently, compute BACC over all window predictions. Primary metric for benchmarking comparison.
- **Per-recording (REC-WIN):** same N=16 evenly-spaced window extraction as probe training, mean-pool, classify pooled vector. Primary metric for clinical deployment evaluation.
- No test-time augmentation. No threshold tuning on the eval split.

4 Combined Best Results (Per-Recording Focus)

Table 3 consolidates the best results from the literature and from this work, filtered to include only per-recording metrics — the clinically meaningful unit of analysis for TUAB (one decision per patient). Per-window results from published SSL models are included for benchmarking context, clearly labelled.

Table 3: Best results on TUAB v3.0.1 evaluation set. Primary focus: per-recording (REC-WIN, one prediction per patient). Per-window (WIN) results from published SSL models included for benchmarking. **Bold** = best in each category. EEGNet and ShallowConvNet are supervised baselines. FEMBA and LaBraM (EEG-Bench) use external large-scale pre-training data.

Method	Encoder	Params	BACC_{WIN}	BACC_{REC}	$\text{AUROC}_{\text{REC}}$	Mode
Supervised baselines (TUAB labels only)						
EEGNet [9]	depthwise CNN	9k	0.796	0.824	0.913	FT
ShallowConvNet [12]	shallow CNN	44k	0.777	0.803	0.893	FT
Deep4Net [12]	deep CNN	–	0.816^{b}	–	–	FT
Foundation models with large external pre-training (published)						
FEMBA-Base [13]	Bidirect. Mamba	47.7M	–	0.810	0.889	FT (TUEG 21kh)
FEMBA-Huge [13]	Bidirect. Mamba	386M	–	**0.818**	**0.901**	FT (TUEG 21kh)
LaBraM-Base [5, 6]	Patch Transformer	5.8M	0.814	0.838	–	FT
SSL models, TUAB-only pre-training (published)						
ContraWR [17]	CNN	–	0.775	–	–	Frozen
BENDR [8]	Conv+Transformer	4M	0.717	–	–	FT
BIOT [18]	FFT Transformer	3.2M	0.802	–	–	Frozen
LaBraM-Base [5]	Patch Transformer	5.8M	**0.814**	–	–	Frozen
EB-JEPA (this work) — SSL frozen (TUAB only)						
VICReg base	Conv1D	0.4M	0.756	0.796	0.888	SSL Frozen
VICReg + corruption	Conv1D	0.4M	0.770	0.825	0.904	SSL Frozen
SIGReg + corruption	Conv1D	0.4M	0.775	0.825	0.904	SSL Frozen
VICReg + spectral $0.1^{\dagger}$	Conv1D	0.4M	0.765	0.836	0.887	SSL Frozen
EB-JEPA (this work) — fine-tuned						
VICReg + corruption → FT	Conv1D	0.4M	–	**0.812**	**0.908**	Fine-tuned

^{b}Accuracy (not BACC); TUAB near-balanced. †Trained with buggy VICReg coefficients (inv_coeff=1 vs. correct 25).

Discussion of Table 3.

- The 0.4M Conv1D encoder matches the 5.8M LaBraM-Base at per-window BACC when using SIGReg + corruption augmentation (both 0.775).
- Per-recording, the fine-tuned encoder reaches BACC 0.812±0.004 (3 seeds, *final* epoch), statistically matching supervised EEGNet (0.812 ± 0.013); the frozen probe reaches 0.819 ± 0.004. SSL pre-training without labels thus *reaches* — but does not exceed — the supervised baseline under patient-level aggregation. (An earlier single-seed, *best-epoch* reading of 0.837 suggested SSL beat EEGNet; that selected the best epoch on the evaluation set, which peeks at the test data, and is retracted — see Lesson 4.)
- The 3.9 pp gap to LaBraM at per-window remains unexplained; the three candidate explanations (tokeniser quality, model scale, pretraining duration) are confounded at this stage.
- AUROC 0.908 (fine-tuned, 3-seed mean) confirms that the representations are high-quality even if the linear separability (frozen probe) is not yet optimal.

5 Conclusion and Future Work

5.1 Summary

We presented EB-JEPA, a two-view energy-based SSL framework for clinical EEG abnormality detection on TUAB. The primary contribution is not a new SOTA number but a *systematic diagnostic account* of what works, what fails, and why.

Positive results. A 0.4M Conv1D encoder pre-trained for 20 epochs on 2 717 unlabelled recordings reaches 0.775 WIN BACC (frozen), matching the published ContraWR result and within 3.9 pp of LaBraM-Base. Corruption augmentation (+time masking + spike injection) is the single most effective ingredient (+0.014 WIN, +0.029 REC). Fine-tuning the pre-trained encoder reaches 0.812 REC BACC (3 seeds, final epoch), matching EEGNet (0.812) with no labels used during pre-training.

Diagnostic findings.

1. **VICReg coefficient bug.** The default `inv_coeff=1.0` was 25× below the paper-correct value. With $(w_2, w_V, w_C) = (1, 1, 1)$ instead of $(25, 25, 1)$, the variance term is underweighted; the encoder converges to a partially collapsed minimum. Diagnostic signature: EEGPT inv_loss 0.103 (buggy, epoch 19) vs. 0.018 (fixed, epoch 11).
2. **Per-window / per-recording evaluation gap.** The framework defaults to REC-WIN (N=16), which is ≈ 5 pp above WIN for the same encoder. Comparing REC to WIN across papers produces a spurious advantage. This caused our initial claim of "beating LaBraM" (0.836 REC vs. 0.814 WIN) to be retracted upon discovering the protocol difference.
3. **EEGPT architectural conflict.** Channel-pool (19 channels → 1 token per time patch) combined with channel-drop augmentation creates cross-view structural incompatibility: the invariance loss L_2 cannot be minimised without learning features that are independent of which channels are present, which degrades the representation for fixed-channel TUAB inference.

5.2 Open Scientific Questions

D1 Is EEGPT collapse due to the architectural conflict, the VICReg bug, or both independently? Jobs with fix-only (no channel-drop change) and fix + no channel-drop are needed to separate the two.

D2 What explains the 3.9 pp gap to LaBraM-Base? Candidates: (i) tokeniser quality (LaBraM uses a VQ-codebook pretrained on Fourier spectra), (ii) model scale (5.8M vs. 0.4M), (iii) pre-training duration (200 vs. 20 epochs), (iv) data volume. Each requires a separate ablation.

D3 LogReg vs. MLP probe: our MLP gives ≈0.01–0.02 BACC higher than LogReg would; switching is needed for direct comparison to SimCLR / BYOL literature.

D4 Does spectral regularisation add independent signal at the correct VICReg coefficients? The current best (0.836 REC) used buggy coefficients; the combination is untested.

D5 Does the aperiodic spectral audit [3] apply to EB-JEPA? The flattening test should be run on our Conv1D encoder to determine whether SIGReg representations capture oscillatory biomarkers or merely the $1/f$ broadband slope.

5.3 Future Work

1. **Neural tokeniser for LaBraM-style encoder.** LaBraM's pretrained VQ-codebook neural spectrum tokeniser (trained via masked patch prediction on Fourier spectra [5])

is the most likely differentiator at 5.8M params. Training this tokeniser separately and plugging into our 1.2M backbone is the highest-priority experiment for closing the gap.

2. **Longer pre-training and larger scale.** All runs used 20 epochs. LaBraM used ≈200 epochs on 2 500h of diverse EEG. FEMBA [13] used 21,000h. Doubling to 40 epochs on TUAB-only is a cheap test of whether capacity or duration limits current performance.
3. **EEGPT with encoder-specific augmentation policy.** Removing channel-drop from EEGPT's augmentation stack (while keeping it for Conv1D and LaBraM) is the minimal fix to resolve the architectural conflict.
4. **SSM-based encoder (Mamba-style).** FEMBA [13] demonstrates that bidirectional Mamba blocks scale linearly in sequence length (vs. $\mathcal{O}(N^2)$ for Transformers), which is attractive for long clinical EEG recordings. Replacing the Conv1D backbone with a small Mamba encoder (≈7.8M, FEMBA-Tiny scale) would test whether the SSM architectural prior improves the energy-based objective.
5. **Aperiodic reliance audit.** Apply Bindra & Panwar's phase-preserving Fourier intervention framework [3] to all our frozen encoders to determine whether EB-JEPA representations exploit the $1/f$ aperiodic shortcut.
6. **Predictive JEPA with per-frame probe.** The EMA-GRU predictive variant (BACC 0.764 REC) was probed via global average pooling, which discards the temporal structure the objective learned. A per-frame temporal probe (one label per time step, aggregated) may recover useful information from predictive representations.

5.4 Methodological Lessons

This project illustrates four recurring failure modes in EEG deep learning:

1. **Comparison without protocol matching.** WIN vs. REC-WIN is the TUAB-specific instance of a general problem: different papers measure different quantities. Always verify the exact evaluation protocol of the paper you compare against before reporting a gap.
2. **Loss coefficient bugs are invisible from convergence curves.** A 25× under-weighted term may still converge to a "good enough" minimum; the bug is only detectable at downstream evaluation or by auditing the raw loss magnitudes against the paper's reported ranges.
3. **Augmentation–architecture coupling.** Augmentation choices that are benign for one architecture can be catastrophic for another (channel-drop + channel-pool). The augmentation stack should be designed or ablated per encoder family, not applied uniformly.
4. **Best-epoch selection peeks at the test set.** Reporting the epoch with the highest *evaluation* accuracy uses the test set for model selection — a subtle leak. Our fine-tuned probe appeared to reach 0.837 REC BACC at its best epoch but only 0.812 ± 0.004 at the final epoch (3 seeds); the 2.5 pp gap is pure peeking, and it was exactly the margin that made SSL look like it *beat* EEGNet rather than tying it. Every model must share one fixed stopping rule (we report final-epoch, seed-averaged); selecting per-model best epochs against the same split you report on manufactures wins that vanish under a held-out protocol.

Code and data. All encoder architectures, training scripts, evaluation protocols, and the science-check protocol are available in the project repository (`Hack_the_worlds/`). The repository includes the EB-JEPA library [14] and all hackathon experiment logs (`reports/EXPERIMENTS.md`, `reports/EVALUATION_ANALYSIS.md`).

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